

CS5187 Vision and Image - Final Mock Exam (Revision Mode)

CS5187 视觉与图像 - 最终模拟试卷 (复习专用版)

Instructions: This document is formatted for revision. Each question is immediately followed by its correct answer and a detailed explanation. 本通过文档专为复习设计。每道题后面紧跟正确答案和详细解析。

Section A: Multiple Choice Questions (10 Questions)

Q1. Image Representation / 图像表示

Question: What is the essential representation of a grayscale image in a computer? 灰度图像在计算机中的本质表示是什么?

A. A continuous analog signal (连续模拟信号) B. A 2D spatially discretized matrix of intensity values (强度的二维空间离散矩阵) C. A set of vector graphics instructions (一组矢量图形指令) D. A 1D array of frequency components (频率分量的一维数组)

Correct Answer (正确答案): B

Explanation (解析):

- **English:** A digital image is sampled (discretized) spatially into pixels, and each pixel represents a quantized intensity value. Thus, it is stored as a 2D matrix (height $\times$ width).
- **Chinese:** 数字图像在空间上被采样（离散化）为像素，每个像素代表一个量化的强度值。因此，它以二维矩阵（高 $\times$ 宽）的形式存储。

Q2. Interpolation Artifacts / 插值伪影

Question: When upsampling an image by 10x using Nearest Neighbor interpolation, what is the most common artifact? 使用最近邻插值将图像放大 10 倍时，最常见的伪影是什么?

A. Motion blur (运动模糊) B. Color distortion (颜色失真) C. Aliasing or "jaggies" (blocky artifacts) (混叠或“锯齿”（块状伪影）) D. Gaussian noise (高斯噪声)

Correct Answer (正确答案): C

Explanation (解析):

- **English:** Nearest Neighbor interpolation simply repeats the value of the closest pixel. At large upscaling factors (like 10x), this makes the individual square pixels clearly visible, creating a blocky, jagged effect known as aliasing.
- **Chinese:** 最近邻插值只是简单地重复最近像素的值。在放大倍数很大（如 10 倍）时，这会使单个方形像素清晰可见，产生称为混叠的块状、锯齿状效果。

Q3. Color Spaces / 色彩空间

Question: Why is the HSI color space often preferred over RGB for image analysis tasks? 为什么在图像分析任务中，HSI 色彩空间通常比 RGB 更受青睐?

A. It is the native format for all camera sensors. (它是所有相机传感器的原生格式) B. It decouples intensity (brightness) from chromaticity (color), making it robust to lighting changes. (它将强度 (亮度) 与色度 (颜色) 解耦, 使其对光照变化具有鲁棒性) C. It uses less memory than RGB. (它比 RGB 占用更少的内存) D. It represents colors using additive mixing of light. (它使用光的加色混合来表示颜色)

Correct Answer (正确答案): B

Explanation (解析):

- **English:** In RGB, changing the brightness affects all three (R, G, B) channels. In HSI (Hue, Saturation, Intensity), the brightness information is isolated in the 'I' channel. We can process color (H, S) without being affected by shadows or uneven lighting.
- **Chinese:** 在 RGB 中, 改变亮度会影响所有三个 (R, G, B) 通道。在 HSI (色调、饱和度、强度) 中, 亮度信息被隔离在 'I' 通道中。我们可以处理颜色 (H, S) 而不受阴影或不均匀光照的影响。

Q4. Filters / 滤波器

Question: Which filter is most effective for removing "Salt-and-Pepper" noise while preserving edges? 哪种滤波器去除“椒盐”噪声同时保留边缘最有效?

A. Mean Filter (均值滤波器) B. Gaussian Filter (高斯滤波器) C. Median Filter (中值滤波器) D. Sobel Filter (Sobel 滤波器)

Correct Answer (正确答案): C

Explanation (解析):

- **English:** Salt-and-pepper noise manifests as random white or black pixels (outliers). Linear filters (Mean/Gaussian) blur these spots. The Median filter (non-linear) replaces the pixel with the median neighbor, effectively removing the outlier without blurring the edges.
- **Chinese:** 椒盐噪声表现为随机的白点或黑点 (离群值)。线性滤波器 (均值/高斯) 会模糊这些点。中值滤波器 (非线性) 用中值邻居替换像素, 在不模糊边缘的情况下有效去除离群值。

Q5. Harris Corner / Harris 角点

Question: In Harris Corner Detection, if the F-value (Noble's measure) is high, what does it imply about the eigenvalues (λ_1, λ_2)? 在 Harris 角点检测中, 如果 F 值 (Noble 度量) 很高, 这暗示特征值 (λ_1, λ_2) 具有什么特征?

A. $\lambda_1 \gg \lambda_2$ (Edge / 边缘) B. $\lambda_1 \approx 0, \lambda_2 \approx 0$ (Flat / 平坦) C. $\lambda_1 \approx \lambda_2$ and both are large (Corner / 角点) D. $\lambda_1 < 0, \lambda_2 < 0$ (Invalid / 无效)

Correct Answer (正确答案): C

Explanation (解析):

- **English:**
 - $\lambda_1 \approx 0, \lambda_2 \approx 0$: Flat region.
 - $\lambda_1 \gg \lambda_2$: Edge.

- λ_1, λ_2 **both large**: Corner. A high F-value (Determinant/Trace) mathematically requires both eigenvalues to be significant.
- **Chinese:**
 - $\lambda_1 \approx 0, \lambda_2 \approx 0$: 平坦区域。
 - $\lambda_1 \gg \lambda_2$: 边缘。
 - λ_1, λ_2 都很大: 角点。高 F 值 (行列式/迹) 在数学上要求两个特征值都显著。

Q6. Gaussian Pyramid / 高斯金字塔

Question: Which of the following parameters is **NOT** required when generating a Gaussian Pyramid?
生成高斯金字塔时, **不需要**以下哪个参数?

A. Number of levels (Octaves) (层数) B. Smoothing parameter σ (平滑参数) C. Subsampling rate (下采样率) D. Camera focal length (相机焦距)

Correct Answer (正确答案): D

Explanation (解析):

- **English:** A Gaussian Pyramid is purely an image processing operation (smoothing + downsampling). It does not depend on the physical properties of the camera (like focal length) that captured the image.
- **Chinese:** 高斯金字塔纯粹是一种图像处理操作 (平滑 + 下采样)。它不依赖于拍摄图像的相机的物理属性 (如焦距)。

Q7. SIFT Invariance / SIFT 不变性

Question: What is the primary mechanism SIFT uses to achieve **rotation invariance**? SIFT 使用什么主要机制来实现**旋转不变性**?

A. Using circular windows for all calculations (在所有计算中使用圆形窗口) B. Assigning a dominant orientation to each keypoint and rotating the descriptor relative to it (为每个关键点分配主方向, 并相对于该方向旋转描述符) C. Ignoring gradient direction and only using magnitude (忽略梯度方向, 只使用幅值) D. Normalizing the image size before processing (在处理前归一化图像大小)

Correct Answer (正确答案): B

Explanation (解析):

- **English:** SIFT computes a histogram of gradient orientations to find the "dominant orientation" of the keypoint. The coordinate system of the descriptor is then rotated to align with this dominant orientation. This ensures the descriptor looks the same regardless of the image's rotation.
- **Chinese:** SIFT 计算梯度方向直方图以找到关键点的“主方向”。然后旋转描述符的坐标系以与该主方向对齐。这确保了无论图像如何旋转, 描述符看起来都是一样的。

Q8. Optical Flow / 光流

Question: In the Lucas-Kanade algorithm, why is a window of pixels used instead of a single pixel? 在 Lucas-Kanade 算法中, 为什么要使用像素窗口而不是单个像素?

A. To reduce computational cost (降低计算成本) B. To solve the Aperture Problem (under-constrained system) (解决孔径问题 (欠约束系统)) C. To increase image resolution (提高图像分辨率) D. To handle large displacements (处理大位移)

Correct Answer (正确答案): B

Explanation (解析):

- **English:** For a single pixel, the optical flow equation ($I_x u + I_y v = -I_t$) has two unknowns (u, v) but only one equation. This is the **Aperture Problem**. Using a window provides multiple equations (one per pixel) for the same (u, v), allowing us to solve it using Least Squares.
- **Chinese:** 对于单个像素，光流方程 ($I_x u + I_y v = -I_t$) 有两个未知数 (u, v) 但只有一个方程。这就是**孔径问题**。使用窗口可以为同一个 (u, v) 提供多个方程 (每个像素一个)，从而允许我们使用最小二乘法求解。

Q9. Homography / 单应性

Question: Which property is preserved under a Homography (Projective Transformation)? 在单应性 (射影变换) 下，哪个属性是保持不变的？

A. Parallelism (平行性) B. Lengths of lines (线段长度) C. Angles (角度) D. Cross-ratio of four collinear points (四个共线点的交比)

Correct Answer (正确答案): D

Explanation (解析):

- **English:**
 - Affine transforms preserve parallelism.
 - Similarity transforms preserve angles.
 - **Projective transforms (Homography)** preserve **Collinearity** (lines remain lines) and the **Cross-Ratio**. Parallel lines may converge (vanishing points).
- **Chinese:**
 - 仿射变换保留平行性。
 - 相似变换保留角度。
 - **射影变换 (单应性)** 保留**共线性** (直线保持直线) 和**交比**。平行线可能会收敛 (消失点)。

Q10. Stereo Vision / 立体视觉

Question: For a stereo vision system, what is the relationship between depth (Z) and disparity (d)? 对于立体视觉系统，深度 (Z) 与视差 (d) 之间的关系是什么？

A. Linear ($Z \propto d$) B. Inverse ($Z \propto 1/d$) C. Quadratic ($Z \propto d^2$) D. Independent (无关系)

Correct Answer (正确答案): B

Explanation (解析):

- **English:** The fundamental stereo equation is $Z = \frac{f \cdot B}{d}$ (where f is focal length, B is baseline). Depth is **inversely proportional** to disparity. Closer objects have larger disparity.
- **Chinese:** 基本立体视觉方程是 $Z = \frac{f \cdot B}{d}$ (其中 f 是焦距, B 是基线)。深度与视差**成反比**。越近的物体视差越大。

Section B: Discussion Questions (4 Questions)

Q11. Edge Detection (边缘检测)

Question: (a) List the four main steps of the Canny Edge Detection algorithm. (b) Explain how **Hysteresis Thresholding** works and why it is better than a single threshold. **问题:** (a) 列出 Canny 边缘检测算法的四个主要步骤。(b) 解释**滞后阈值**是如何工作的, 以及为什么它比单一阈值更好。

Answer & Explanation: (a) Four Steps:

1. **Gaussian Smoothing:** Filter the image to remove noise. (高斯平滑: 过滤图像以去除噪声)
2. **Gradient Calculation:** Compute magnitude and direction of gradients (Sobel). (梯度计算: 计算梯度的幅值和方向)
3. **Non-Maximum Suppression:** Thin edges to 1-pixel width by suppressing non-peak values. (非极大值抑制: 通过抑制非峰值将边缘细化为1像素宽)
4. **Hysteresis Thresholding:** Detect and connect edges using two thresholds. (滞后阈值: 使用两个阈值检测并连接边缘)

(b) Hysteresis Mechanism:

- **Method:** Use a High threshold (T_{high}) and a Low threshold (T_{low}).
 - Pixels $> T_{high}$ are "Strong Edges" (definitely edge).
 - Pixels between T_{low} and T_{high} are "Weak Edges".
 - **Key Rule:** Weak edges are kept **only if** they are connected to a Strong Edge.
- **Why Better:** A single threshold is too rigid. If too high, edges break (gaps). If too low, noise creates false edges. Hysteresis bridges gaps in valid edges while ignoring isolated noise.
- **Chinese:**
 - **方法:** 使用高阈值 (T_{high}) 和低阈值 (T_{low})。
 - 大于 T_{high} 的像素是“强边缘”(确定是边缘)。
 - 介于 T_{low} 和 T_{high} 之间的像素是“弱边缘”。
 - **关键规则:** 弱边缘只有在连接到强边缘时才保留。
 - **为什么更好:** 单一阈值太死板。太高会导致边缘断裂; 太低会产生噪声伪边缘。滞后阈值在忽略孤立噪声的同时, 修补了有效边缘的断裂。

Q12. RANSAC Calculation (RANSAC 计算)

Question: You are using RANSAC to fit a model.

- Outlier ratio (ϵ) = 50% (0.5).

- Sample size (s) = 4 points.
- Desired success probability (P) = 99% (0.99). Calculate the minimum number of iterations (N) required. Show your formula. 问题：你正在使用 RANSAC 拟合模型。离群率 $\epsilon = 0.5$ ，样本大小 $s = 4$ ，目标成功率 $P = 0.99$ 。计算所需的最少迭代次数 N 。展示你的公式。

Answer & Explanation: 1. Identify Parameters:

- Probability of selecting an inlier in one pick: $w = 1 - \epsilon = 1 - 0.5 = 0.5$.
- Probability of selecting a "clean" sample set (all 4 inliers): $w^s = 0.5^4 = 0.0625$.
- Probability of selecting a "bad" sample set (at least one outlier):
 $1 - w^s = 1 - 0.0625 = 0.9375$.

2. Formula:

$$N = \frac{\log(1 - P)}{\log(1 - w^s)}$$

3. Calculation:

$$N = \frac{\log(1 - 0.99)}{\log(0.9375)} = \frac{\log(0.01)}{\log(0.9375)} \approx \frac{-2}{-0.028} \approx 71.4$$

4. Conclusion: Round up to the nearest integer. $N = 72$ iterations. (向上取整，需要 72 次迭代。)

Q13. Image Transformation (图像变换)

Question: You want to transform a poster on a wall to an image plane using a Homography matrix H .

(a) How many degrees of freedom does H have? (b) What is the minimum number of corresponding point pairs needed? (c) If you have more points than needed, what method would you use to solve for H ? 问题：你想使用单应性矩阵 H 将墙上的海报变换到图像平面。(a) H 有多少个自由度？(b) 最少需要多少对对应点？(c) 点数多于最少需求时怎么解？

Answer & Explanation: (a) Degrees of Freedom (DOF): 8. The homography matrix is 3×3 (9 elements), but it is scale-invariant (multiplying by a constant doesn't change the transformation). So $9 - 1 = 8$ DOF. (8个自由度。3x3矩阵有9个元素，但具有尺度不变性，减去1个。)

(b) Min Point Pairs: 4 pairs. Each point pair $(x, y) \rightarrow (u, v)$ provides 2 independent linear equations (one for x , one for y). Since we have 8 unknowns, we need $8 \div 2 = 4$ points. (4对点。每对点提供2个方程，8个未知数需要4个点。)

(c) Method for Extra Points: Least Squares Estimation (or DLT - Direct Linear Transformation). Since the system is over-determined (more equations than unknowns), we find an approximate solution that minimizes the reprojection error. (最小二乘法 或 DLT。因为系统是超定的，我们寻找最小化误差的近似解。)

Q14. Optical Flow (光流)

Question: (a) List the three key assumptions of the Lucas-Kanade optical flow method. (b) What happens if the "Small Motion" assumption is violated? How can we modify the algorithm to handle this?

问题：(a) 列出 Lucas-Kanade 光流方法的三个关键假设。(b) 如果“小运动”假设被违反会发生什么？如何修改算法？

Answer & Explanation: (a) Three Assumptions:

1. **Brightness Constancy:** The intensity of a pixel stays constant as it moves from frame t to $t + 1$. (亮度恒定)
2. **Small Motion:** The displacement is small enough (sub-pixel) to justify the Taylor series linearization. (小运动：位移足够小以适用泰勒级数线性化)
3. **Spatial Coherence:** Pixels in a small local neighborhood window move with the same velocity. (空间一致性：局部窗口内的像素速度相同)

(b) Violation & Solution:

- **Problem:** If motion is large, the linear Taylor approximation ($I(x + u) \approx I(x) + I_x u$) is no longer valid, and the algorithm fails to converge.
- **Solution:** Use a **Pyramid (Coarse-to-Fine)** approach.
 1. Build a Gaussian pyramid for the images.
 2. Estimate flow at the coarsest level (where large motion becomes small motion).
 3. Upsample the flow and warp the image, then refine at finer levels.
- **Chinese:**
 - **问题：** 如果运动较大，泰勒线性近似失效，算法无法收敛。
 - **解法：** 使用**金字塔（由粗到细）**方法。在最粗糙的层级（大运动变小运动）估计光流，然后上采样并细化。